·Solutions·

1.

1. When all rays just reach the sphere, the edge ray is tangent at T (Fig. 1); geometry gives

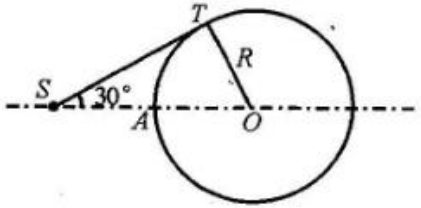


Fig. 1

$$\sin 30^\circ = \frac{R}{\overline{SO}}$$

Hence

$$\overline{SA} = R$$

If the source is over R from A, some rays miss.

The whole beam reaches the cap if source - A distance obeys

$$x \leqslant R \tag{1}$$

2. For $x = R$, a ray at α to the axis: Fig. 2.

B is on the axis; dashed QBQ' and QAQ' form a sphere. θ and θ' are incidence/refraction angles; intersection: P'.

β: OP' - axis angle; α': refracted ray - axis angle.

The sine rule for $\triangle OPS$ gives

$$\frac{2R}{\sin\theta} = \frac{R}{\sin\alpha} \tag{2}$$

The refraction law gives

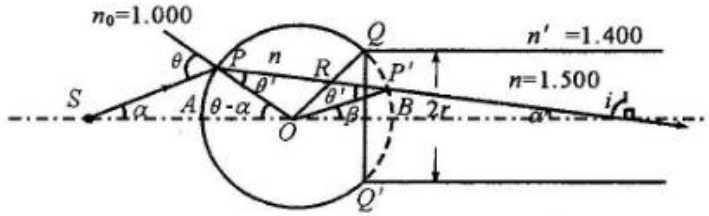


Fig. 2

$$\sin\theta = 1.5\sin\theta' \tag{3}$$

Geometry gives

$$\beta = 2\theta' - \theta + \alpha \tag{4}$$

$$\alpha' = \theta - \alpha - \theta' \tag{5}$$

If its backward extension meets the axis: Fig. 3. Set the sign of α' as follows: a ray bent towards the axis gives

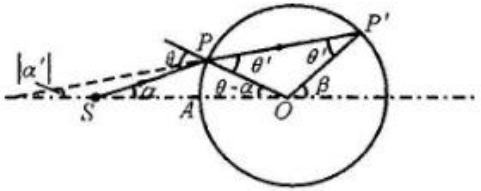


Fig. 3

positive α'. If it bends away (the backward extension meets the axis left of A), α' is negative; equation (5) still holds.

Geometry also gives the fibre-entry condition:

$$|\beta| \leqslant \angle QOB \tag{6}$$

β may be negative, meaning P' is below the axis.

and $$\angle QOB = \arcsin\frac{r}{R}$$

With data, fibre entry is equivalent to

$$|\beta| \leqslant 33.75^\circ$$

Consider the three rays in turn.

(i) $\alpha_1 = 30^\circ$:

a. Can the refracted ray enter the fibre?

Using (2), $\sin\theta_1 = 2\sin\alpha_1 = 1$; hence

$$\theta_1 = 90^\circ$$

Substitute into (3):

$$\theta_1' = \arcsin\left(\frac{2}{3}\right) = 41.81^\circ$$

Substitute into (4):

$$\beta_1 = 2\theta_1' - \theta_1 + \alpha_1 = 23.62^\circ \qquad (7)$$

Since

$$\beta_1 < \angle QOB \qquad (8)$$

the ray enters the fibre.

b. Can total internal reflection occur at the core - cladding interface? Let the critical angle be i_C, then

$$i_C = \arcsin\left(\frac{1.400}{1.500}\right) = 68.96^\circ \qquad (9)$$

By $i = 90^\circ - \alpha'$, total reflection requires α' to satisfy the following:

$$\alpha' \leqslant 90^\circ - 68.96^\circ = 21.04^\circ \qquad (10)$$

Now $\alpha_1' = \theta_1 - \alpha_1 - \theta_1' = 18.19^\circ$ meets (10); thus it propagates by total internal reflection in the fibre.

(ii) $\alpha_2 = 25^\circ$:

Using (2): $\sin\theta_2 = 2\sin\alpha_2 = 2\sin25^\circ$ $\quad \theta_2 = 57.70^\circ$

$$\sin\theta_2' = \frac{1}{1.5}\sin\theta_2 \qquad \theta_2' = 34.30^\circ$$

$$\beta_2 = 2\theta_2' - \theta_2 + \alpha_2 = 35.90^\circ \qquad (11)$$

(6): ray cannot enter the fibre right of QQ'.

(iii) $\alpha_3 = 20^\circ$:

Using (2): $\sin\theta_3 = 2\sin\alpha_3 = 2\sin20^\circ$ $\quad \theta_3 = 43.16^\circ$

$$\sin\theta_3' = \frac{1}{1.5}\sin\theta_3 \qquad \theta_3' = 27.13^\circ$$

$$\beta_3 = 2\theta_3' - \theta_3 + \alpha_3 = 31.10^\circ \qquad (12)$$

(6): ray enters the fibre right of QQ'.

Then

$$\alpha_3' = \theta_3 - \alpha_3 + \theta_3' = -3.97^\circ \qquad (13)$$

Negative: backward ray meets the axis left of A, hitting the upper boundary from below (Fig. 4). $|\alpha_3'| < 21.04^\circ$; it can propagate in the fibre by total internal reflection.

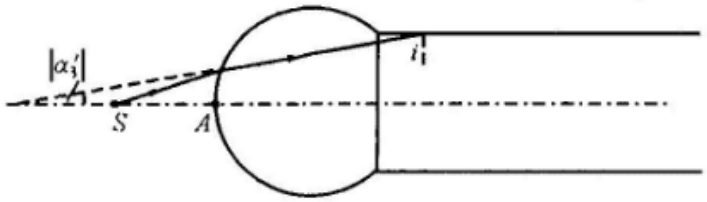


Fig. 4

2.

1. Electron rest mass m_0, photon frequency ν; assume it absorbs the photon completely and moves at v; momentum conservation then gives

$$\frac{h\nu}{c} = \frac{m_0 v}{\sqrt{1-(v^2/c^2)}} \tag{1}$$

The total energy after collision is

$$E = mc^2 = \frac{m_0 c^2}{\sqrt{1-(v^2/c^2)}} \tag{2}$$

Eliminate v using (1)、(2):

$$E = \sqrt{m_0^2 c^4 + h^2\nu^2} \tag{3}$$

Electron - photon total energy before collision:

$$E_0 = h\nu + m_0 c^2 \tag{4}$$

Using (3)、(4),

$$E^2 - E_0^2 = (m_0^2 c^4 + h^2\nu^2) - (h\nu + m_0 c^2)^2 =$$

$$-2h\nu m_0 c^2 \neq 0 \tag{5}$$

This violates energy conservation, so the supposed process cannot occur.

2. Metal-bound electron + incident photon: momentum is not conserved in photoemission. Use energy conversion.

Escape from metal needs at least work W (work function). After escape: speed v, photon energy $h\nu$, electron rest mass m_0. Photoemission requires

$$h\nu \geqslant \left[\frac{m_0 c^2}{\sqrt{1-(v^2/c^2)}} - m_0 c^2\right] + W \tag{6}$$

The emitted speed v is much less than light speed; hence

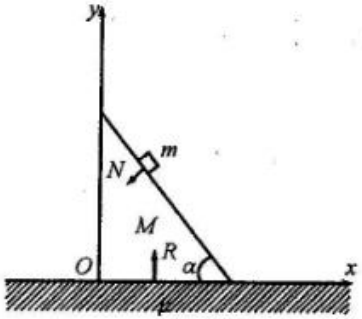
$$\frac{1}{\sqrt{1-(v^2/c^2)}} \approx 1 + \frac{1}{2}\frac{v^2}{c^2} + \cdots \tag{7}$$

Drop higher terms; substitute the first two of (7) in (6):

$$h\nu \geqslant \frac{1}{2} m_0 v^2 + W \tag{8}$$

Thus $h\nu \geqslant W$ suffices for (8) and photoemission.

3. For ease, consider the small block on BC. Oxy as shown; a_x、a_y are x, y acceleration magnitudes. N, R: block - wedge and wedge - table normal forces. Let a_M denote the wedge acceleration magnitude. Block equations:



$$ma_x = N\sin\alpha \tag{1}$$

$$ma_y = mg - N\cos\alpha \tag{2}$$

Wedge equations:

$$Ma_M = N\sin\alpha - \mu R \tag{3}$$

$$R = N\cos\alpha + Mg \tag{4}$$

The block is constrained to the slope:

$$\frac{a_x + a_M}{a_y} = \cot\alpha \tag{5}$$

Solving the above equations gives

$$N = \frac{(\cos\alpha + \mu\sin\alpha)mg}{1 + (m/M)\sin^2\alpha - \mu(m/M)\sin\alpha\cos\alpha} \tag{6}$$

On AB, replace α by β. Now decide whether the wedge moves, using (3). If it does, then (3) and (4) require

$$a_M = \frac{1}{M}(N\sin\alpha - \mu N\cos\alpha - \mu Mg) > 0 \tag{7}$$

Else the wedge stays still; the block slides down it.

On BC, $\alpha = 45°$, using (6)、(7) and the given data yields

$$N = 28.2\ \text{N} \tag{8}$$

$$a_M = 2.50 \times 10^{-2}\ \text{m}\cdot\text{s}^{-2} > 0 \tag{9}$$

On AB, use $\beta = 60°$ in (6)、(7) in place of $\alpha = 45°$; N' for N, a_M' for a_M. From (6)、(7) and numerical data,

$$N' = 20.1\ \text{N} \tag{8'}$$

$$a_M' = -3.90 \times 10^{-2}\ \text{m}\cdot\text{s}^{-2} < 0 \tag{9'}$$

Thus wedge rests from A to B, but moves from B to C. The block's work on the wedge is

$$W = N\sin\alpha \cdot s \tag{10}$$

s is the wedge displacement;

$$s = \frac{1}{2}a_M t^2 \tag{11}$$

Find t from vertical block motion from B to C, using (2) and (8):

$$a_y = g - \frac{N}{m}\cos\alpha = 5.01\ \text{m}\cdot\text{s}^{-2}$$

By kinematics,

$$h_2 = v_{0y}t + \frac{1}{2}a_y t^2 \tag{12}$$

v_{0y}: vertical part of v_0 just past B. Obtain v_0 by energy conservation:

$$\frac{1}{2}mv_0^2 = mg(h_1 - h_2) \tag{13}$$

Thus
$$v_0 = \sqrt{2g(h_1 - h_2)} = 4\ \text{m}\cdot\text{s}^{-1}$$
$$v_{0y} = v_0\cos\alpha \tag{14}$$

Use (14) in (12) to find

$$t = \frac{-v_{0y} \pm \sqrt{v_{0y}^2 + 2a_y h_2}}{a_y}$$

Take the positive sign and insert data:

$$t = 0.492\ \text{s}$$

Use (11), recalling (9):

$$s = \frac{1}{2}a_M t^2 = 3.01 \times 10^{-3}\ \text{m} \tag{15}$$

Use (10):

$$W = N \cdot \sin\alpha \cdot s = 6.00 \times 10^{-2}\ \text{J} \tag{16}$$

4.

Right of A、B, 8 capacitors (Fig. 1) equal one C_{AB}. Simplify to Fig. 2. First find C_{AB}.

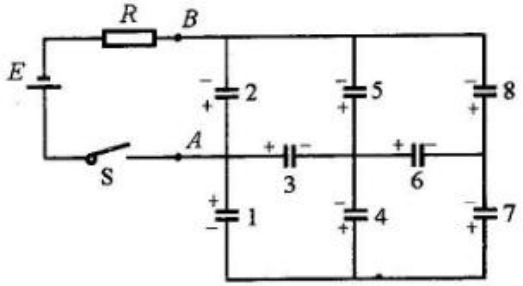


Fig. 1

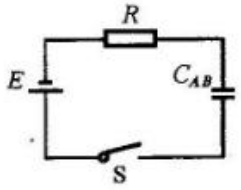


Fig. 2

S shut: final charge from A、B to right of A、B is Q (A positive, B negative). Capacitor i: q_i; plate charge signs are as in Fig. 1.

Node charge balance: 4 independent equations:

$$q_1 + q_2 + q_3 = Q \tag{1}$$

$$-q_1 + q_4 + q_7 = 0 \tag{2}$$

$$-q_3 - q_4 + q_5 + q_6 = 0 \tag{3}$$

$$-q_6 - q_7 + q_8 = 0 \tag{4}$$

Loop voltages, e.g. $\frac{q_1}{C} + \frac{q_4}{C} - \frac{q_3}{C} = 0$, give

$$q_1 + q_4 - q_3 = 0 \tag{5}$$

$$q_4 + q_6 - q_7 = 0 \tag{6}$$

$$q_2 - q_3 - q_5 = 0 \tag{7}$$

$$q_5 - q_6 - q_8 = 0 \tag{8}$$

Both cap. 2 and equivalent C_{AB} have voltage E; hence

$$E = \frac{q_2}{C} \tag{9}$$

$$E = \frac{Q}{C_{AB}} \tag{10}$$

Solving (1) ~ (8) gives

$$q_2 = \frac{8}{15}Q \tag{11}$$

From (9)、(10)、(11),

$$C_{AB} = \frac{15}{8}C \tag{12}$$

During charging in Fig. 2, total circuit charge is Q; the source does work

$$W_0 = EQ \tag{13}$$

Energy stored in the equivalent capacitor:

$$W_1 = \frac{1}{2}QE \tag{14}$$

Energy conservation: heat dissipated in the resistor is

$$W_2 = W_0 - W_1 \tag{15}$$

From the above equations,

$$W_2 = \frac{15}{16}E^2C \tag{16}$$

5.

1. **Particle speeds on entering the magnetic field**

At $t = 0$ open D_1. Since $u = U_0\sin\left(\frac{2\pi}{T}t\right)$, C_1's voltage is $u = 0$, so all particles pass C_1. After C_1, if they reach C_2 at $t = T/2$ or integer multiples of $T/2$, C_2's voltage is zero, so they pass C_2. To pass C_1 then C_2, their speeds v_n must satisfy

$$\frac{l}{v_n} = n\frac{T}{2} \quad n = 1,2,\cdots \tag{1}$$

For particles passing C_1 at $t = 0$ then C_2, speeds are

$$v_n = \frac{2l}{nT} \quad n = 1,2,\cdots \tag{2}$$

Time from C_1 to C_2 at speed v_n:

$$\Delta t_{n1} = \frac{1}{2}nT \tag{3}$$

Time from C_2 to G:

$$\Delta t_{n2} = \frac{l}{2v_n} \tag{4}$$

Thus arrival time at G is

$$t_n = \Delta t_{n1} + \Delta t_{n2} = \frac{3}{4}nT \tag{5}$$

Since D_2 shuts at $t = 10T/4$, only if

$$t_n \leqslant \frac{10}{4}T \tag{6}$$

they pass C into the field. Using (5)、(6), n is an integer, so the largest n is

$$n_{max} = 3 \tag{7}$$

Their energies on entering the magnetic field are

$$E_n = \frac{1}{2}mv_n^2 + E_0\sin\left[\frac{2\pi}{T}\left(\frac{3n}{4}T\right) + \frac{\pi}{4}\right]$$
$$n = 1,2,3 \tag{8}$$

Their energies are, respectively,

$$E_1 = \frac{7}{4}\frac{ml^2}{T^2} \qquad E_2 = \frac{1}{4}\frac{ml^2}{T^2} \qquad E_3 = \frac{17}{36}\frac{ml^2}{T^2} \tag{9}$$

$t = T/2$, C_1 at $u = 0$: all speeds again pass C_1. Those with speed v_n' satisfying

$$\frac{l}{v_n'} = \frac{n}{2}T \qquad n = 1,2,\cdots \tag{10}$$

can also pass C_2. Their speeds are

$$v_n' = \frac{2l}{nT} \qquad n = 1,2,\cdots \tag{11}$$

Times from C_1 to C_2 and C_2 to G, respectively:

$$\Delta t_{n1}' = \frac{n}{2}T \qquad \Delta t_{n2}' = \frac{n}{4}T \tag{12}$$

Time of arrival at G:

$$t_n' = \frac{T}{2} + \Delta t_{n1}' + \Delta t_{n2}' = \frac{3n+2}{4}T \tag{13}$$

At $t = 10T/4$, D_2 shuts, hence

$$t_n' \leqslant \frac{10}{4}T \tag{14}$$

By (13) and (14), n being integral,

$$n_{max} = 2 \tag{15}$$

These particles enter the field with energies

$$E_n' = \frac{1}{2}mv_2'^2 + E_0\sin\left[\left(\frac{2\pi}{T}\frac{3n+2}{4}\right)T + \frac{\pi}{4}\right]$$
$$n = 1,2 \tag{16}$$

Entry energies are, respectively,

$$E_1' = \frac{9}{4}\frac{ml^2}{T^2} \qquad E_2' = \frac{3}{4}\frac{ml^2}{T^2} \tag{17}$$

At $t = 3T/4$, D_1 shuts: no later entry to C_1. By (9), (17), max entry energy is E_1', giving greatest speed V_1'. Using $E_1' = \frac{1}{2}MV_1'^2$,

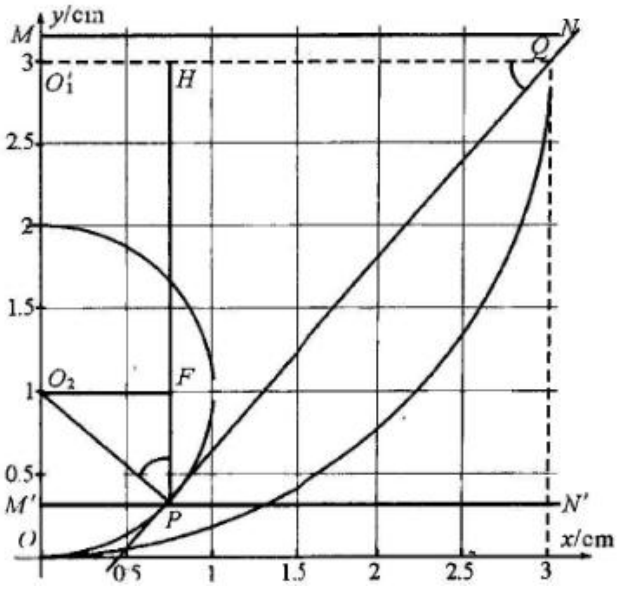
$$V_1' = \sqrt{\frac{9}{2}}\frac{l}{T} \tag{18}$$

Least energy E_2 gives least speed V_2; similarly, we get

$$V_2 = \sqrt{\frac{1}{2}}\frac{l}{T} \tag{19}$$

2. Determine the magnetic-field boundary

In the shown Cartesian field frame, entering at O,



The fastest particle passes Q at (3 cm,3 cm). By (18)、(19), greatest entry speed V_1' is 3 times least V_2; their circular radii R_1' and R_2 in the field obey

$$R_1' = 3R_2$$

The circle for speed V_1' passes origin O, tangent to x, and passes Q, giving

$$R_1' = 3 \text{ cm}$$

Center O_1' is at (0 cm,3 cm). The slower orbit has radius

$$R_2 = 1 \text{ cm}$$

O_2 lies at (0 cm,1 cm); its circle is

$$x^2 + (y-1)^2 = 1 \tag{20}$$

Draw a tangent from Q, touching P at (x_P, y_P). If it exits the field at P with V_2, it follows PQ to Q. P must lie on the field boundary.

Find P. O_2P is normal to PQ. Drop P onto $O_1'Q$ at H; drop O_2 onto PH at F. Triangles $\triangle O_2PF$ and $\triangle PQH$ are similar, so

$$\frac{\overline{O_2F}}{\overline{PF}} = \frac{\overline{PH}}{\overline{HQ}} \tag{21}$$

i.e.

$$\frac{x_P}{(1-y_P)} = \frac{(3-y_P)}{(3-x_P)} \tag{22}$$

giving $$x_P^2 + y_P^2 - 4y_P = 3x_P - 3$$

Completing the square:

$$x_P^2 + (y_P-1)^2 - 2y_P - 1 = 3x_P - 3$$

Point P at coordinates (x_P, y_P) satisfies (20),

$$-2y_P = 3x_P - 3$$

$$y_P = \frac{3}{2}(1 - x_P) \tag{23}$$

Using (20), the y coordinate of P is

$$y_P = 0.354 \text{ cm} \tag{24}$$

Thus shift boundary MN down parallel through P to $M'N'$ as shown. At V_2 it exits at P along the circle tangent to reach Q.

6.

As a、b move right, b and c first meet and collide. Impact is brief: both springs retain their natural lengths. a、d are unchanged; b、c have equal mass. Elastic impact leaves b at rest and c at v_0, rightward. Just after impact, a has v_0 rightward; b rests. Mass center M_1 of a and b is the spring midpoint, at $v_0/2$ rightward; c has v_0 rightward, d rests. For c and d, mass center M_2 is also the spring midpoint, at $v_0/2$ rightward.

Origin M_1 follows the mass center rightward at $v_0/2$. Its X' axis joins all four centers (mass-center frame). Take the first b and c impact as the time

origin ($t = 0$). Here the a、b spring midpoint is fixed. a、b form two fixed-end spring oscillators: natural length $l/2$, stiffness $2k_1$. Likewise c、d form two oscillators, each of natural length $l/2$ and stiffness $2k_2$.

For a、b、c、d, x_a'、x_b'、x_c'、x_d' are displacements from their own equilibria. Then

$$x_a' = A_1\cos(\omega_1 t + \varphi_1)$$
$$x_b' = A_1\cos(\omega_1 t + \varphi_2)$$

where $\omega_1 = \sqrt{\dfrac{2k_1}{m}}$

A_1、φ_1、φ_2 are constants to determine.

$$x_c' = A_2\cos(\omega_2 t + \varphi_3)$$
$$x_d' = A_2\cos(\omega_2 t + \varphi_4)$$

where $\omega_2 = \sqrt{\dfrac{2k_2}{m}}$

A_2、φ_3、φ_4 are constants to determine.

At $t = 0$, relative to the center-of-mass frame,

$$x_{a0}' = 0 \qquad v_{a0}' = \frac{v_0}{2}$$
$$x_{b0}' = 0 \qquad v_{b0}' = -\frac{v_0}{2}$$
$$x_{c0}' = 0 \qquad v_{c0}' = \frac{v_0}{2}$$
$$x_{d0}' = 0 \qquad v_{d0}' = -\frac{v_0}{2}$$

These are the initial conditions.

Reference circles give four center-frame velocities, respectively:

$$v_a' = -A_1\omega_1\sin(\omega_1 t + \varphi_1)$$
$$v_b' = -A_1\omega_1\sin(\omega_1 t + \varphi_2)$$
$$v_c' = -A_2\omega_2\sin(\omega_2 t + \varphi_3)$$
$$v_d' = -A_2\omega_2\sin(\omega_2 t + \varphi_4)$$

Displacements, velocities and initial conditions give

$$A_1 = \frac{v_0}{2\omega_1} \qquad \varphi_1 = -\frac{\pi}{2} \qquad \varphi_2 = \frac{\pi}{2}$$
$$A_2 = \frac{v_0}{2\omega_2} \qquad \varphi_3 = -\frac{\pi}{2} \qquad \varphi_4 = \frac{\pi}{2}$$

Hence

$$\left.\begin{aligned} x_a' &= \frac{v_0}{2\omega_1}\sin\omega_1 t \\ x_b' &= -\frac{v_0}{2\omega_1}\sin\omega_1 t \\ x_c' &= \frac{v_0}{2\omega_2}\sin\omega_2 t \\ x_d' &= -\frac{v_0}{2\omega_2}\sin\omega_2 t \end{aligned}\right\} \qquad (1)$$

$$\left.\begin{aligned} v_a' &= \frac{v_0}{2}\cos\omega_1 t \\ v_b' &= -\frac{v_0}{2}\cos\omega_1 t \\ v_c' &= \frac{v_0}{2}\cos\omega_2 t \\ v_d' &= -\frac{v_0}{2}\cos\omega_2 t \end{aligned}\right\} \qquad (2)$$

The four center-of-mass-frame coordinates are

$$\left.\begin{aligned} X_a' &= -\frac{l}{2} + \frac{v_0}{2\omega_1}\sin\omega_1 t \\ X_b' &= \frac{l}{2} - \frac{v_0}{2\omega_1}\sin\omega_1 t \\ X_c' &= \frac{l}{2} + \frac{v_0}{2\omega_2}\sin\omega_2 t \\ X_d' &= \frac{3l}{2} - \frac{v_0}{2\omega_2}\sin\omega_2 t \end{aligned}\right\} \qquad (3)$$

Use a table-fixed frame: its origin is the position of the ab mass center M_1 just after impact. Align X with X'. The four table-frame coordinates are

$$\left.\begin{aligned} X_a &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_1}\sin\omega_1 t - \frac{l}{2} \\ X_b &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_1}\sin\omega_1 t + \frac{l}{2} \\ X_c &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_2}\sin\omega_2 t + \frac{l}{2} \\ X_d &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_2}\sin\omega_2 t + \frac{3l}{2} \end{aligned}\right\} \qquad (4)$$

The block velocities relative to the table are

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_1 t \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_1 t \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_2 t \\ v_d &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_2 t \end{aligned}\right\} \qquad (5)$$

1. For $k_1 = k_2$

If both springs have stiffness k, then

$$\omega_1 = \omega_2 = \sqrt{\frac{2k}{m}} = \omega_0 \qquad (6)$$

After first impact, use (4) for a、b、c、d and (5) for their table positions and velocities.

$$\left.\begin{aligned} X_a &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_0}\sin\omega_0 t - \frac{l}{2} \\ X_b &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_0}\sin\omega_0 t + \frac{l}{2} \\ X_c &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_0}\sin\omega_0 t + \frac{l}{2} \\ X_d &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_0}\sin\omega_0 t + \frac{3l}{2} \end{aligned}\right\} \quad (7)$$

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_0 t \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_0 t \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_0 t \\ v_d &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_0 t \end{aligned}\right\} \quad (8)$$

If b、c meet again (the second time), then

$$X_b = X_c$$

From (7),

$$\sin\omega_0 t = 0 \quad (9)$$

Least positive nonzero t:

$$\omega_0 t = \pi \quad (10)$$

b、c center-frame velocities at meeting, by (2), (9):

$$v_b'(\text{at}) = \frac{v_0}{2}$$

$$v_c'(\text{at}) = -\frac{v_0}{2}$$

b、c meet: b right, c left; b、c collide again.

Just after collision they exchange velocities:

$$v_b'(\text{aft}) = -\frac{v_0}{2}$$

$$v_c'(\text{aft}) = \frac{v_0}{2}$$

Second b、c meeting: a、b、c、d shifts from equilibria follow from (10) and (1):

$$x_a' = 0$$

$$x_b' = 0$$

$$x_c' = 0$$

$$x_d' = 0$$

Thus both springs are natural and undeformed.

After second impact, table velocities of a、b、c、d:

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \\ v_d &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \end{aligned}\right\} \qquad (11)$$

After second impact, a、b rest; springs are natural. At their equilibria, a、b rest and c、d move right together at v_0, uniformly, without further oscillations or collisions.

2. For $k_1 = 4k_2$

Since $k_1 = 4k_2$,

$$\omega_1 = 2\omega_2 = \sqrt{\frac{2k_1}{m}} = \sqrt{\frac{8k_2}{m}} \qquad (12)$$

For a、b、c、d, after first impact, (4) and (5) give the table-frame positions and velocities.

$$\left.\begin{aligned} X_a &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_1}\sin\omega_1 t - \frac{l}{2} \\ X_b &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_1}\sin\omega_1 t + \frac{l}{2} \\ X_c &= \frac{1}{2}v_0 t + \frac{v_0}{\omega_1}\sin\frac{\omega_1 t}{2} + \frac{l}{2} \\ X_d &= \frac{1}{2}v_0 t - \frac{v_0}{\omega_1}\sin\frac{\omega_1 t}{2} + \frac{3l}{2} \end{aligned}\right\} \qquad (13)$$

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_1 t \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_1 t \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2}\cos\frac{\omega_1 t}{2} \\ v_d &= \frac{v_0}{2} - \frac{v_0}{2}\cos\frac{\omega_1 t}{2} \end{aligned}\right\} \qquad (14)$$

If b and c meet in their later motion, then

$$X_b = X_c$$

From (13),

$$-\frac{1}{2}\sin\omega_1 t = \sin\left(\frac{\omega_1 t}{2}\right)$$

Since $\omega_1 = 2\omega_2$,

$$\sin\omega_2 t(\cos\omega_2 t + 1) = 0 \qquad (15)$$

The roots are

$$\omega_2 t = n\pi \qquad n = 0, \pm 1, \pm 2, \cdots \qquad (16)$$

The least positive nonzero solution t satisfies

$$\omega_2 t = \pi \qquad (17)$$

and

$$\omega_1 t = 2\pi \qquad (18)$$

b、c center-frame velocities at meeting, by (2), (16):

$$v_b'(\text{at}) = -\frac{v_0}{2}$$

$$v_c'(\text{at}) = -\frac{v_0}{2}$$

b、c meet at equal speeds leftward: b、c do not collide again. At b、c's second meeting, (17)、(18), (1) give the four displacements from their own equilibria.

$$x_a' = 0$$
$$x_b' = 0$$
$$x_c' = 0$$
$$x_d' = 0$$

Both springs are natural. The four table velocities are, respectively,

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_c &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_d &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \end{aligned}\right\} \qquad (19)$$

a right, b still: compression speeds up b, slows a. c rests, d moves: stretch speeds up c, slows d. The spring oscillators resume oscillating.

From (16), when

$$\omega_2 t = 2\pi$$
$$\omega_1 t = 4\pi$$

b、c meet again: b、c center-frame velocities by (2) are

$$v_b'(\text{at}) = -\frac{v_0}{2}$$
$$v_c'(\text{at}) = \frac{v_0}{2}$$

b left, c right, equally fast: b、c meet without impact. The four table velocities are

$$\left.\begin{aligned} v_a &= v_0 \\ v_b &= 0 \\ v_c &= v_0 \\ v_d &= 0 \end{aligned}\right\} \qquad (20)$$

Then a、b mass center M_1 moves right uniformly at $\frac{1}{2}v_0$.

a、b move harmonically about M_1 at angular frequency ω_1. c、d have mass center M_2. It moves uniformly right at $\frac{1}{2}v_0$. c、d do so about M_2 at angular frequency ω_2. b、c meet every $\frac{\pi}{\omega_2}$ time units, always in either case above: meeting without colliding.